

ICCT COLLEGES FOUNDATION INC

College on Engineering and Digital Technology

PROJECT NEXUS

Community Aid Network (CAN): A Web-Based Platform for Disaster Response and Social Support

A Capstone Project Presented to the Faculty of the
College of Engineering and Digital Technology, in
Partial Fulfillment of the Requirements for the Course
Software Design

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Approval Sheet

This project, titled "Community Aid Network (CAN)", prepared and submitted by; Art Lenard Badrecina, Jose Mariano Ladrera, Argie Etac, Eunice Gay Bayani, Adrian Montes, Ara Jean Canatoy, Lanze Axhel Kenneth Cu, Shanti Quiban, and Anjho Tuvilla, in partial fulfillment of the requirements for the course Software Design, has been examined and is recommended for acceptance and approval.

Adviser

Accepted and approved in partial fulfillment of the requirements for the degree of Master of Arts in Education major in Physical Education and Sports.

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Acknowledgment

[Optional. A brief note from the team thanking advisers, panel members, and contributors. One page maximum.]

CHAPTER 1 – THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Communities such as barangays, shelters, and orphanages often face serious challenges when disasters like typhoons, fires, or poverty strike. Based on observations and conversations with local leaders, the current process of asking for help is scattered — requests are shared through social media posts, word of mouth, or informal notices. This leads to delays, confusion, and sometimes duplication of aid, making it harder for essential goods and medicines to reach those who need them most.

To address this gap, the study proposes a web-based application that serves as a centralized platform for verified community representatives to post their needs and for helpers to find and respond directly. Unlike donation systems focused on money, this project emphasizes essential items and medical supplies, ensuring that tangible resources are prioritized. By streamlining communication and improving transparency, the platform aims to strengthen disaster response, support vulnerable groups, and promote a culture of shared responsibility within the community.

1.2 Statement of the Problem

Communities affected by disasters, poverty, and other social challenges often have difficulty effectively communicating their needs and connecting with individuals willing to provide assistance. At the same time, donors and volunteers may struggle to find reliable, relevant, and up-to-date opportunities to provide aid, resulting in inefficient and ineffective coordination and delayed assistance.

Specifically, our study aims to address the following problems:

1. **Unorganized Requests**

   Problem: Aid requests are currently spread across social media, word of mouth, or informal notices, leading to confusion and delays.

   System Feature: A centralized posting system where verified community representatives can submit structured requests for essentials.

2. **Difficulty in Finding Communities in Need**

   Problem: Donors and volunteers cannot easily identify which communities require urgent help.

   System Feature: A searchable database with filters (location, type of need, urgency) to help helpers quickly find communities to support.

3. **Lack of Direct Communication**

   Problem: Helpers often have no reliable way to coordinate with community leaders, causing miscommunication or duplication of aid.

System Feature: A secure messaging system that connects helpers directly with barangay officials or community representatives.

4. **Limited Transparency in Aid Distribution**

   Problem: Donors may not know if their contributions reached the intended recipients, reducing trust and willingness to help.

   System Feature: A request tracking and update mechanism where communities can mark fulfilled needs and provide progress updates.

5. **Constraints in Accessibility**

   Problem: Communities in remote areas with poor internet access may face challenges in using the system.

   System Feature: A lightweight, mobile-friendly interface designed to work on basic smartphones and low-bandwidth connections.

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this project is to develop a centralized and accessible web-based application that connects communities in need — such as barangays affected by disasters, orphanages, animal shelters, and other vulnerable groups — with individuals and organizations willing to provide assistance. By integrating a structured system for posting requests, searching for communities, and facilitating direct communication between helpers and recipients, the project aims to enhance the efficiency, transparency, and timeliness of community aid distribution. Ultimately, it seeks to strengthen resilience, promote social responsibility, and create a sustainable platform for mutual support across diverse communities.

1.3.2 Specific Objectives

1. Organized Requests - To establish a centralized posting system where verified barangay officials and community leaders can submit structured requests for essential goods and medicines, ensuring that at least 80% of requests are posted through the platform within the first four months.
2. Finding Communities in Need - To develop a searchable database with filters for location, urgency, and type of need, enabling helpers to quickly identify communities, with at least 100 registered communities accessible by the sixth month.
3. Direct Communication - To integrate a secure messaging feature that connects helpers directly with community representatives, facilitating at least 200 successful exchanges within six months and reducing miscommunication by 50%.
4. Transparency in Aid Distribution - To implement a request tracking and update mechanism where communities can mark fulfilled needs, ensuring that by the end of the first year, at least 70% of posted requests include status updates for accountability.

5. Accessibility Constraints - To design a lightweight, mobile-friendly interface that functions on basic smartphones and low-bandwidth connections, demonstrating usability during pilot testing in three barangays within twelve months.

1.4 Scope and Limitation

The scope of this study centers on the development of a web-based application that serves as a bridge between communities in need and individuals willing to provide assistance. Specifically, the system is designed to support barangays, orphanages, animal shelters, and other vulnerable groups by allowing verified representatives to post requests for essential goods and medicines. Helpers and volunteers can search for these requests, filter them by urgency or location, and directly communicate with the community leaders to coordinate the delivery of aid. By focusing on essentials rather than financial donations, the platform ensures that tangible, life-sustaining resources reach those who need them most. The system also includes features such as request tracking, secure messaging, and basic transparency mechanisms to build trust between communities and helpers.

However, the project is limited in several important ways. The system does not handle monetary transactions or financial contributions, restricting its scope to physical goods and medical supplies. Its effectiveness depends heavily on the accuracy of information provided by community representatives, as the platform cannot independently verify the actual delivery of donations. Technical constraints such as internet connectivity and access to smartphones or computers may also limit participation, especially in remote areas. Additionally, the project will operate within a fixed timeframe and budget, focusing first on core features and pilot testing in selected barangays before expanding to a wider scale. These limitations highlight the need for careful planning, realistic expectations, and gradual improvement as the system evolves.

1.5 Significance of the Study

This study is significant because it introduces a web-based application that directly improves how communities request and receive essential goods and medicines. By creating a centralized, transparent, and accessible system, the project ensures that aid reaches those who need it most, while also fostering stronger collaboration between communities and helpers.

Direct Beneficiaries:

- Communities in Need (barangays, shelters, orphanages, animal welfare groups): Gain a structured way to request and receive essential goods and medicines, reducing delays and confusion during emergencies.
- Community Leaders and Representatives: Benefit from an organized system to post verified requests, track fulfillment, and coordinate with helpers, minimizing duplication of aid and logistical errors.
- Helpers and Volunteers: Have a reliable platform to identify urgent needs, connect directly with communities, and provide assistance with confidence that their contributions are reaching the intended recipients.

Indirect Beneficiaries:

- Local Communities and Barangays – Receive structured support during disasters and emergencies, improving resilience and recovery.
- Donors and Volunteers – Benefit from a reliable platform that assures their contributions reach verified recipients, encouraging continued participation in community aid.
- Society at Large – Gains from a culture of solidarity and shared responsibility, where technology bridges the gap between those in need and those willing to help.

1.6 Definition of Terms

Accessibility – In this study, accessibility refers to the ability of users, especially those in remote areas with limited internet or basic smartphones, to effectively use the system without technical barriers.

Aid Distribution – The process of delivering essential goods and medicines from helpers to communities, which the system supports through request tracking and updates.

Barangay Officials / Community Representatives – Verified leaders authorized to post requests for their community's needs within the platform, ensuring legitimacy and accountability.

Centralized Posting System – A structured feature of the application where all requests for help are submitted in one place, reducing scattered communication and delays.

Community Aid Network (CAN) – The proposed web-based application developed in this study, designed to connect communities in need with helpers and volunteers.

Database – The organized collection of community requests and helper information, searchable by filters such as location, urgency, and type of need.

Donors / Helpers / Volunteers – Individuals or groups willing to provide assistance in the form of essential goods and medicines, not financial support.

Messaging System – A secure communication feature within the app that allows helpers and community representatives to coordinate aid directly.

Request Tracking – A transparency mechanism where communities update the status of posted needs, marking them as fulfilled or ongoing to build trust.

Transparency – The practice of ensuring accountability by showing updates on requests and aid received, fostering confidence between communities and helpers.